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Roll No

EC-8004 (3) (CBGS)
B.E. VIII Semester Examination, June 2020
Choice Based Grading System (CBGS)
Radar Engineering
Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) Derive the radar range equation.
b) Discuss the working of multiple frequency radar.
2. a) Explain the principle of Doppler effect. And also explain the working of CW radar.
b) Explain the working of FM CW altimeter.
3. a) Discuss the principle working of MTI Radar processor.
b) Discuss about delay line Canceller and Doppler filter.
4. a) Discuss about Radar cross section.
b) What is the effect of polarization on cross section?
5. a) Explain about real radar signals and complex radar signals.
b) Discuss the working of matched filter in respect to Radar application.
6. a) Explain the concept of Ambiguity Function.
b) Discuss about the transmission of signals through networks.
7. a) Discuss about radar antenna and radar resolution.
b) Explain the principle of PPI representation displays.
8. Write short notes on any two of the following.
 - a) Multiple or staggered pulse repetition
 - b) FMCW Radar
 - c) Scattering cross section
